

Time: 70 min

Max marks: 20

Note: solve any four questions. all questions carry equal marks. Make suitable assumptions whenever required.

- Q.1** Design full subtractor using two half subtractors circuit and explain its working. 5
- Q.2** What are the advantages offered by an edge triggered RS flip flop over a clocked or gated RS flip flop? 5
- Q.3** Design a SISO shift register using JK flip flop and also draw the waveform to shift the binary number 1101 into the register for four clock transitions. 5
- Q.4** Design 3-bit synchronous up counter using D flip flops sensitive to clock NT. 5
- Q.5** The four variable function F is given in terms of min-terms as 5
 $f(a, b, c, d) = \sum m(2, 3, 5, 10, 11, 12, 13, 15)$. Using the K map Minimize the function in the SOP form. Also give the realization using only two input NAND gates.